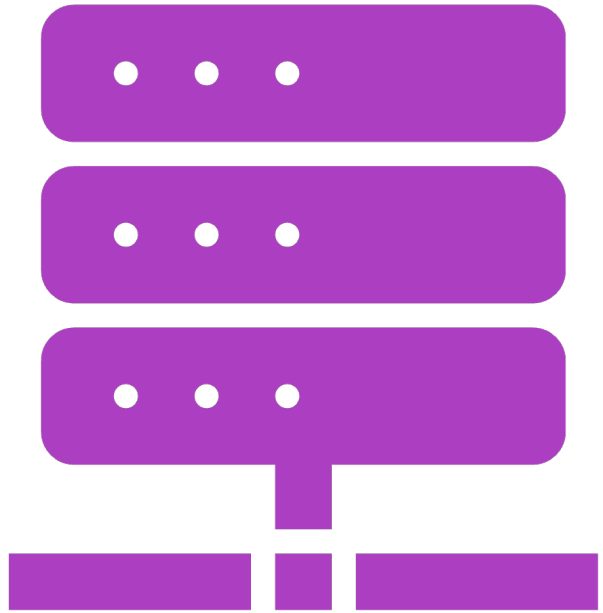
DESIGN AND SIMULATION OF A MULTI-LAYER SECURE ENTERPRISE NETWORK

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INTRODUCTION



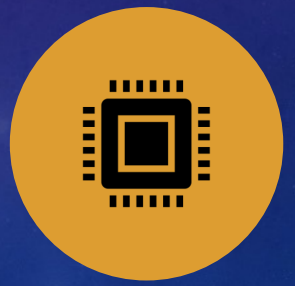
Organizations rely heavily on computer networks for varied operations. Their network becomes increasingly vulnerable.



Traditionally networks often focus primarily on connectivity. These networks may not provide adequate security.



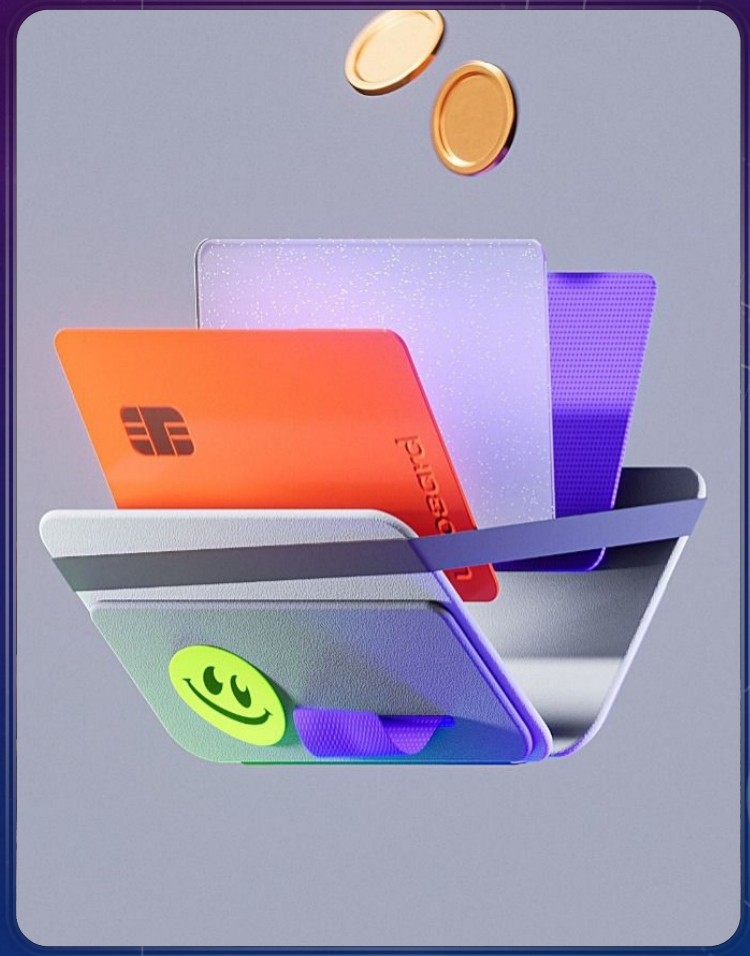
A secure and scalable network architecture becomes essential and necessary.



The project’s aim is to design and simulate and validate such secure enterprise network.

COMPANY BACKGROUND SECUREPAY TECHNOLOGIES PVT. LTD.

- SecurePay Technologies is a fictional FinTech company providing:
  - Online payment processing
  - Digital Financing transactions
  - Customer account management
  - Financial data storage



DEPARTMENTS



HUMAN RESOURCES



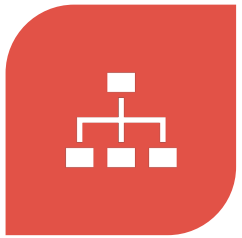
FINANCE



IT SUPPORT



CUSTOMER SERVICE



ADMINISTRATION



GUEST USERS

PROBLEM STATEMENT

Organizations may use poorly segmented networks where multiple departments share common resources. Such environments are vulnerable to unauthorized access, threats, etc.

Hence, there is a need for implementation and evaluation of a secure network architecture capable of:

- Detecting and preventing unauthorized access attempts.
- Protecting critical organizational resources.
- Implementing multiple layers of network defense.
- Evaluating the effectiveness of different Cisco security mechanisms.
- Ensuring secure and reliable client and server communication.





EXISTING SYSTEM 1 TRADITIONAL FLAT ENTERPRISE NETWORK

Description: All devices are connected within the same network without logical separation.

Reference: Cisco Enterprise Campus Design Guide

Advantages:

- Simple deployment
- Low cost

Limitations:

- No network segmentation
- Easy lateral movement for attackers
- High security risk

EXISTING SYSTEM 2 VLAN BASED ENTERPRISE NETWORK

Description: Departments are separated using VLANs.

Reference: Cisco VLAN Design Documentation

Advantage:

- Better Segmentation
- Reduced broadcast traffic

Limitations:

- Does not prevent rogue devices
- Additional security mechanisms required

EXISTING SYSTEM 3 ACL-BASED SECURITY NETWORKS

Description: Access control lists filter traffic between departments.

Reference: Cisco ACL Configuration Guide

Advantage:

- Better access control
- Traffic filtering

Limitations:

- Complex rule management
- Human configuration errors



EXISTING SYSTEM 4 PORT SECURITY NETWORKS

Description: Restricts access to switch ports by allowing only authorized devices.

Reference: Cisco Catalyst Security Guide

Advantages:

- Prevents unauthorized device connections
- Improves endpoint security

Limitations:

- Protects only switch ports
- Cannot stop network-wide attacks



EXISTING SYSTEM 5
DHCP SNOOPING NETWORKS

Description: Monitors DHCP traffic and prevents rogue DHCP servers.

Reference: Cisco DHCP Snooping Documentation

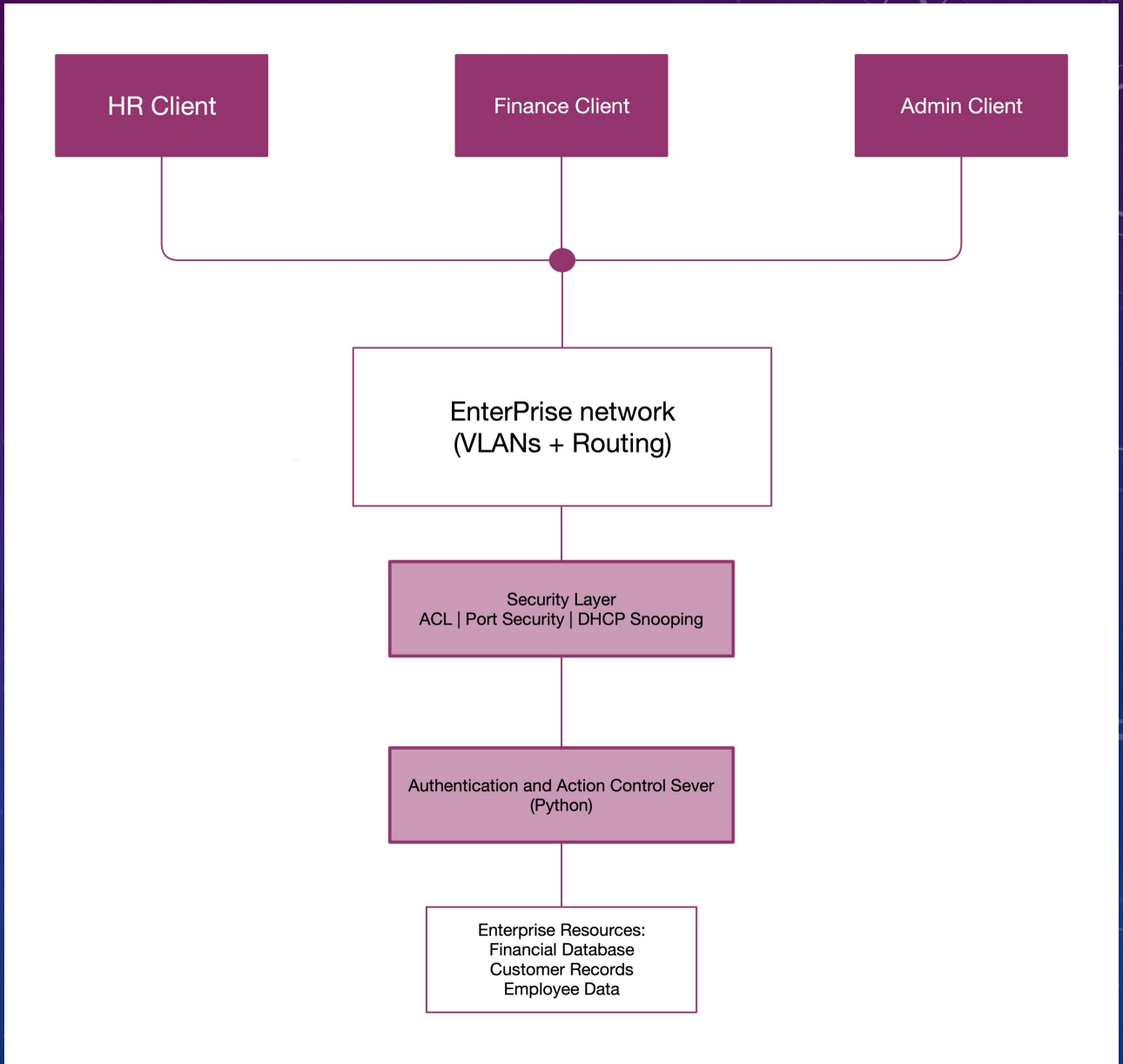
Advantages:

- Protects against DHCP spoofing

Limitations

- Protects only one attack vector
- Additional security layers still required

PROPOSED ARCHITECTURE



GAP ANALYSIS

Limitations of Existing Systems

- Only individual security control
- No single approach
- Gaps still exist between departments
- Requries multiple coordinated security mechanisms
- Effectiveness of security controls is rarely validated through attack simulations

Security Mechanism	Limitations
VLAN	Segmentation only
ACL	Rule Complexity
Port Security	Localized protections
DHCP snooping	Limited scope
Flat Network	No Security

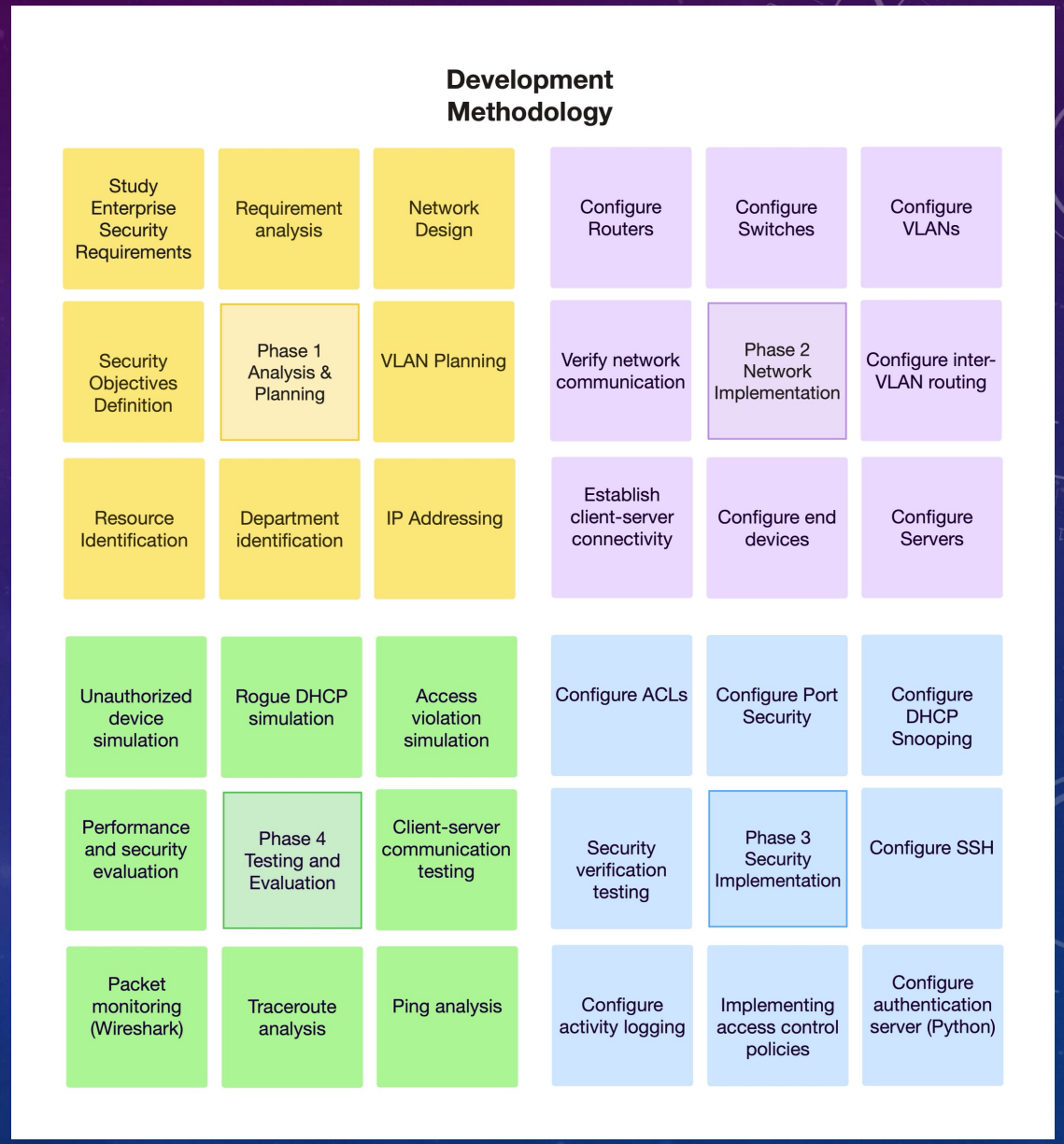
NOVELTY

Existing solutions focus on individual security mechanisms, the proposed system integrates multiple Cisco security technologies and emphasizes on detecting and preventing unauthorized access through a layered security architecture. We validate their effectiveness through simulated attack scenarios and unauthorized access attempts.

Additional Features

- Department-wise isolation
- Role-based access control
- Threat simulation
- Security validation testing
- Client-server communication testing

DEVELOPMENT METHODOLOGY



CLIENT-SERVER ARCHITECTURE

Process

1. Clients sends request
2. Server validates request
3. Access control applied
4. Server sends response

Implemented using Python Socket Programming. Python will also be implemented for user authentication, access control and activity logging.

TOOLS USED

TOOL	PURPOSE
Cisco Packet Tracer	Network Simulation
Python	Client-Server Programming, Authentication, Access Control, and Activity Logging
Wireshark	Packet Analysis
Visual Studio Code	Development Environment



REFERENCES

- Cisco Networking Academy
- Cisco Security Documentation
- Cisco Packet Tracer Documentation
- Kurose & Ross – Computer Networking
- Forouzan – Data Communications and Networking
- OMADA networks
- Tech Target
- CrowdSec
- CBTNuggets